

Exercise 03: Multi-Head Self-Attention

A numerical two-head self-attention example with independent projections.

Setup

Use the row-token input

$$X = \begin{matrix} & \begin{matrix} 2 & & 3 \\ 1 & 0 & 1 & 0 \end{matrix} \\ \begin{matrix} 4 \\ 0 \\ 1 \end{matrix} & \begin{matrix} 1 & 1 & 15 & 2 \end{matrix} \end{matrix} \mathbb{R}^{3 \times 4}.$$

There are $H = 2$ heads with $d_k = d_v = 2$ per head. The projection matrices are fixed toy values; nothing is trained in this exercise.

Head 1

Head 1 uses

$$W_Q^{(1)} = W_K^{(1)} = W_V^{(1)} = \begin{matrix} & \begin{matrix} 2 & & 3 \\ 1 & 0 \\ 6 & 17 \\ 4 & 05 \\ 0 & 0 \end{matrix} \end{matrix}.$$

Therefore

$$Q_1 = K_1 = V_1 = \begin{matrix} & \begin{matrix} 2 & & 3 \\ 1 & 0 \\ 4 & 15 \\ 1 & 1 \end{matrix} \end{matrix}.$$

The scaled score matrix is

$$\frac{Q_1 K_1^T}{\sqrt{2}} = \begin{matrix} & \begin{matrix} 2 & & 3 \\ 0.7071 & 0 & 0.7071 \\ 4 & 0 & 0.70715 \\ 0.7071 & 0.7071 & 1.4142 \end{matrix} \end{matrix}.$$

Applying row-wise softmax gives

$$A_1 = \begin{matrix} & \begin{matrix} 2 & & 3 \\ 0.4011 & 0.1978 & 0.4011 \\ 0.1978 & 0.4011 & 0.40115 \\ 0.2483 & 0.2483 & 0.5035 \end{matrix} \end{matrix}.$$

The first head output is

$$O_1 = A_1 V_1 = \begin{matrix} & \begin{matrix} 2 & & 3 \\ 0.8022 & 0.5989 \\ 0.5989 & 0.80225 \\ 0.7517 & 0.7517 \end{matrix} \end{matrix}.$$

Head 2

Head 2 receives the same X but uses different projections:

$$W_Q^{(2)} = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 0 & 7 \\ 4 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}; \quad W_K^{(2)} = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 0 & 7 \\ 4 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}; \quad W_V^{(2)} = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 0 & 7 \\ 4 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

This gives

$$Q_2 = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 0 & 7 \\ 4 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}; \quad K_2 = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 0 & 7 \\ 4 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}; \quad V_2 = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 0 & 7 \\ 4 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

The scaled score matrix is

$$\frac{Q_2 K_2^T}{\sqrt{2}} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 0.7071 & 1.4142 & 0.7071 \\ 0.7071 & 0.7071 & 0 & 0 \end{bmatrix}$$

Hence

$$A_2 = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 0.7071 & 1.4142 \\ 0.7071 & 0.7071 & 0 \end{bmatrix}$$

and

$$O_2 = A_2 V_2 = \begin{bmatrix} 2 & 0.5989 & 0.8022 \\ 4 & 0.7517 & 0.7517 \\ 0.8022 & 0.5989 & 0 \end{bmatrix}$$

Why the heads differ

The two heads receive the same input matrix X , but they do not see the same projected geometry. Their independent W_Q , W_K , and W_V matrices produce different query-key dot products and therefore different attention distributions:

$$A_1 \neq A_2:$$

This controlled example demonstrates only that independent projections can produce distinct attention patterns. The projection matrices are hand-designed rather than learned, so the exercise does *not* assign semantic roles such as "relational" or "anchoring" to either head.

Concatenation

Concatenate the head outputs along the feature dimension:

$$O_{\text{concat}} = \text{Concat}(O_1; O_2) = \begin{bmatrix} 2 & 0.8022 & 0.5989 & 0.5989 & 0.8022 \\ 4 & 0.5989 & 0.8022 & 0.7517 & 0.7517 \\ 0.7517 & 0.7517 & 0.8022 & 0.5989 & 0 \end{bmatrix}$$

Thus

$$O_{\text{concat}} \in \mathbb{R}^{3 \times 4}.$$

Output projection

For this toy exercise,

$$W_O = I_4:$$

Therefore

$$O_{\text{MHA}} = O_{\text{concat}} W_O = O_{\text{concat}}:$$

The identity choice is only a simplification that keeps the concatenated features visible. In a trained Transformer, W_O is learned and mixes information across head features.

Causal extension

If the same two-head layer were causal, the same causal mask M would be added inside each head before softmax:

$$A_{h,\text{causal}} = \text{softmax} \left(\frac{Q_h K_h^T}{d_k} + M \right):$$

The visibility constraint is shared, while each head still has its own projected logits.

Pipeline summary

For each head h ,

$$Q_h = X W_Q^{(h)}; \quad K_h = X W_K^{(h)}; \quad V_h = X W_V^{(h)};$$

$$\text{head}_h = \text{softmax} \left(\frac{Q_h K_h^T}{d_k} \right) V_h:$$

Then

$$O_{\text{MHA}} = \text{Concat}(\text{head}_1; \dots; \text{head}_H) W_O:$$

Numerical verification

The companion `answer_key.py` reproduces the two attention matrices, both head outputs, the concatenation, and the final identity output projection with deterministic assertions.